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University of Central Punjab

Faculty of Information Technology

Mid Term Exam (CISE 4151): Human Computer Interaction

Question No.	Answering Time	Percent Weight	Total Marks
1-10	10 min	10%	10

Name _____

Reg. _____

INSTRUCTIONS

- This is a **CLOSED BOOK** and **CLOSED NOTES** exam. Cheat sheet is **NOT ALLOWED**.
- Attach this question paper with your answer sheet.
- There are a total of 94 questions, and 82 printed sides of a page.

Q1. Multiple Choice Questions (Provide Answers on Answer Sheet)

- The four main components for an interaction to happen are as follows:
 - ☒ The system, user, input, output
 - The user, computer, human, task
 - Input, task, output, human
 - None
- The factors in the physical environment that directly affect the quality of the interaction and the user's performance are:
 - ☒ Physical position, Time
 - Temperature, lightning
 - Noise
 - All of the above
- What could be some interface styles between computer and humans?
 - Command line interface (CLI)
 - Menus
 - Point and click
 - ☒ All of the above
- Which of the following is one of the constraints that designer of an interaction system can face?
 - Piracy
 - Bugs
 - ☒ Legal and safety
 - None
- For a product to be successful, it must be:
 - Useful by humans
 - Usable by humans
 - Used by humans
 - ☒ All of the above
- Which of the following statement is true regarding personas?
 - ☒ A primary persona needs their own interface
 - Multiple secondary personas are combined into primary persona.
 - A persona represents the elastic user.
 - None of the above
- A computer expert produces a solution with HCI which is very efficient in computer resources, bas command-lines. Which of the following is most likely to be the result when the system is implements
 - Most staff will want to become computer experts to use it.
 - All staff will enjoy using it after mastering the command lines.
 - It will be welcomed by all staff.
 - ☒ Most staff will not want to use the system.
- _____ refers to how good a system at doing what it is supposed to do?
 - Learnability
 - Usefulness

3:04 PM

University of Central Punjab
Faculty of Information Technology
Final Term Exam (CSSE4153): Human Computer Interaction

Duration	Semester	Percent Weight	Total Marks
2h 30m	Spring 2024	40%	50

Name: _____

Reg # _____

INSTRUCTIONS

1. Attach this question paper with your answer sheet.
2. There is a total of 08 questions, and 03 printed page.

Q1. Answer the following questions briefly. Do Not use more than 2-3 sentences [13]

1. What is a mile high menu, explain with an example?
2. Explain mapping with the help of an example?
3. Write names of six usability attributes (names only)?
4. What is the difference between perceived affordance and real affordance?
5. A prototype which is drawn on paper by hand is an example of low fidelity or high fidelity? Justify your choice shortly.
6. Can we have medium fidelity prototypes? Explain your answer shortly.
7. State four things that are provided to evaluators during a briefing session in Heuristic Evaluation?
8. Differentiate between throw-away and evolutionary prototypes.
9. What is the core difference between formative and summative evaluation?
10. Differentiate between recognition and recall.
11. What is the difference between internal and external consistency. Highlight with an example.
12. How inspections are different from testing?
13. What is co-experience with respect to primary and secondary users of a system?

Q2. What are four phases of usability evaluations in product lifecycle (for example, during design, or after completion). Explain how they differ from each other with the help of an example. [4]

Q3. Write any four components of mental model. Explain with a single sentence [4]

Q4. In their new model, Tesla has introduced several new features, including some of the following [4]

- 1- Parental control for cars
- 2- New apps integrated into interface, like YouTube and Netflix
- 3- An improved camera coverage with 360-degree view
- 4- Restricted driving profiles that (dis)allow certain driving modes, for example no sports mode to a person under 22 or over 60.

Discuss each of the mentioned improvement and give your view on what type of innovation it is. Justify with only one sentence if you think its incremental or radical.

CS CamScanner

Q5. In the recent, Pakistani universities have started automating their human-resource related operations, especially that are related to leaves and attendances. The universities prefer that all their employees, including staff members, administration and faculty members shall apply leaves, and maintain their records via online system. [12]

When a leave is applied, the immediate manager (boss of the applying person) reviews the leave request and approves or disapproves. If the leave is approved, it is communicated with the HR department, and is deducted from the quota of that employee. If the leave is disapproved, the manager needs to provide a reason for disapproval, which is displayed back to the requesting employee.

The HR department keeps track of all leaves, and quotas, and prepares each employee's salary of that month accordingly. However, the record is not limited to HR, and managers, Administration, Top Management, and the employee themselves can see the records, though according to their privileges. For example, top management and HR can see everyone's record, but a manager can see record of their team only. Finally, an employee can only see their own record.

According to the given scenario, answer the following questions

- 1- Identify user classes of the described system and mark them as primary / secondary

[2]

Q4: CL01

Marks[5+2]

Evaluate the following design if you find any inconsistency or bad designing then enlist it (any five) and justify your points with reason.

The screenshot displays the Airblue website interface. At the top, the 'airblue' logo is on the left, and navigation links for 'Reservations', 'Travel Deals', 'Destinations', 'Schedule', and 'About Us' are on the right. Below the logo is a 'Search Flights' section with a 'Round Trip' button selected. The 'from' field is set to 'Dhaka, BD' and the 'to' field is 'London, GB'. The 'Departing' date is '14-07-2020' and the 'Returning' date is '21-07-2020'. There are fields for 'Passengers' (1 Adult) and 'Cabin Class' (Economy). A 'SEARCH' button is at the bottom of the search form. To the right of the search form is a large black box with white text titled 'IMPORTANT NOTICE FOR UAE TRAVEL'. The notice states that as per UAE authorities, certain testing is required for passengers to enter the UAE and lists approved COVID-19 testing facilities. At the bottom of the page, there are links for 'bluenews' and 'enjoy onboard'.

airblue Reservations Travel Deals Destinations Schedule About Us

Search Flights

Round Trip One Way

from

to

Departing

Returning

Passengers

Cabin Class

SEARCH

IMPORTANT NOTICE FOR UAE TRAVEL

As per UAE authorities, certain testing is required for passengers to enter the UAE and some at the time of flight.

COVID-19 negative test reports are accepted from following designated labs only:

PERMITTED & APPROVED

International Diagnostic Center
Genex Niyati Lab
Chughtai Lab

NOT PERMITTED

International Diagnostic Center
Genex Niyati Lab
Chughtai Lab
Shaukat Khanum Memorial Trust
Alfa Khan University Hospital

bluenews enjoy onboard

3:07 PM

University of Central Punjab

Faculty of Information Technology

Duration	Semester	Percent Weight	Total Marks
2h 30m	Fall 2023	40%	40

Name _____

Reg # _____

INSTRUCTIONS

1. This is a **CLOSED BOOK** and **CLOSED NOTES** exam.
2. Attach this question paper with your answer sheet.
3. There is a total of 7 questions and 4 printed pages.
4. **CALCULATOR SHARING IS NOT ALLOWED**
5. Providing multiple solutions to a question is not allowed. If multiple answers are found, only the first uncrossed will be checked, and the remaining will be ignored.

Part 1: Short Questions: (Don't use more than a few sentences to answer)

Q1. Answer the following questions.

[6]

- I. What are the measurable attributes of system's rejectability?
- II. Differentiate between novelty and meaningfulness.
- III. How Storyboards are different from Sketches?
- IV. Differentiate between throw-away and evolutionary prototypes.
- V. Differentiate between recognition and recall.
- VI. Differentiate between pre-attentive and attentive processing?

Part 2: Scenario-based Questions:

Read the following scenario carefully, you need to understand it to answer questions in this section.

Imagine you are tasked with designing a health and wellness mobile application namely Healthy Pakistan aimed at promoting a healthier lifestyle for users (both males and females). The app is intended to provide personalized guidance on nutrition, exercise, and mental well-being. Users should be able to set goals, track their progress, and receive real-time feedback. You are also planning to launch the website as well as a mobile application that shall work as follows:

1. The customer visits the Healthy Pakistan website or opens the mobile app.
2. The system presents the customer with the option to log in or order as a guest.

- ☒ c. Effectiveness
 9. Persona is not context specific, so it can be reused easily.
 a. True
☒ b. False
10. Which of the following statement is true regarding the measurement of usability attributes?
 a. Learnability is measured by performing common tasks of a software
☒ b. Learnability is determined by measuring the time it takes to verbally explain an interface to a new user
 c. Expert users are needed to measure Learnability
 d. None of these
2. Answer the following questions shortly.
 a. Differentiate between social acceptability and practical acceptability of a system?
 b. Why human Computer interaction is important in software design?
 c. Explain briefly why personas are useful for design.
 d. What are observation problems with respect to the Abowd and Beale's interaction model?
- A modern high-end hotel in Pakistan has installed a Sandwich Making and Delivery Robot (SMDR) to make and supply room service sandwiches at any hour of the day. The hotel also has an automated delivery system as part of SMDR that will take the completed sandwich to a specified room. The hotel mainly targets foreigners who travel to Pakistan for their business or leisure trips. However, a few people in the wealthy Pakistani community who can afford high-end services are also welcomed.
1. Identify target market of SMDR system.
 2. Identify user classes, and mention the type of each class (i.e., primary, or secondary).
 3. Write one persona for some primary user class you have identified in task 2.
 4. Write a scenario describing how a guest order a sandwich using your interface.

3. The customer selects the desired option and proceeds.
4. If logging in, the customer enters their username and password and clicks "Sign in."
5. The system authenticates the customer's credentials and redirects them to the menu page.
6. The system presents the customer with choices for age, height, and weights, trainer type, exercise day per week, fitness level, health condition, schedules
7. The customer selects the desired options and adds the goals to their cart.
8. The system displays the customer's cart with the selected goals and provides the detail guidance on nutrition, exercise and mental well-being.
9. The customer reviews their cart and clicks "Proceed to Finalize."
10. If the customer is a guest, they are prompted to get registered and pay.
11. The customer enters their address, phone number, and any special instructions.
12. The system calculates the total amount, including taxes, and displays it for the customer to review.
13. The customer verifies the selected details, applies any available coupons or discounts, and clicks "Finalize."
14. The system processes the order, charges the customer's preferred payment method, and generates an order confirmation.
15. The customer receives a confirmation message on the website or app, along with an email or SMS confirmation.
16. The system notifies the nearest personalized trainer about the customer's finalized plan for preparation and guidance.
17. After 30-40 minutes, the trainer arrives for the first session, collects payment if applicable, and hands over the detailed plan to the customer.
18. The customer confirms the plan is accurate and of satisfactory quality.

Sara saw a Healthy Pakistan advertisement about 20% off in an email sent by her university. She wants to give it a try, so, she downloads the Healthy Pakistan app, registers herself, selects a female trainer and health plan and applies the coupon to get a discount.

- | | | |
|-----|--|-----|
| Q2. | Identify the target market / audience of Healthy Pakistan. And classify them into primary or secondary classes. | [2] |
| Q3. | Write a user scenario in which Sara successfully uses Healthy Pakistan mobile app to select a female trainer for daily exercise. | [6] |
| Q4. | Draw a storyboard for Sara that shows how she uses Healthy Pakistan mobile app to provide a female trainer for daily exercise | [6] |

Part 3: Long Questions:

- | | | |
|-----|--|-----|
| Q5. | Run heuristic evaluation through the interface given below. Use Nielsen's heuristics to find any usability problem (you must find at least five problems). Make a spreadsheet and rate the problem on the given severity index. Nielsen's heuristics and severity index are given below. | [5] |
|-----|--|-----|

3:05 PM

Q2: CLO1+CLO2

Marks [2*4=8]

Answer the following questions shortly:

- a. Explain why context is important when evaluating usability.
- b. An elderly person still uses a basic keypad phone instead of a smartphone. Classify the user type and explain your reasoning.
- c. A persona named "Ali" is a first-year student with high anxiety about exams. Suggest features that your app design include based on this persona?
- d. Explain the significance of user-centered design in the development of modern interfaces.

Q3: CLO2

Marks [5+5=10]

Banking software's in Pakistan need to connect with a central server via the internet at each transaction request. That is why, in case of an internet or communication failure, the system of the bank branch cannot process any transactions and valuable time is lost.

1. Suggest five users of the existing system, who should be consulted for the requirements of the new system and why? (Just 1 line)
2. List the questions (main part) you would prepare to interview the aforementioned users (3 from each), for any two users.

9. What is the core difference between formative and summative evaluation?
10. Differentiate between recognition and recall.
11. What is the difference between internal and external consistency. Highlight with an example.
12. How inspections are different from testing?
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According to the given scenario, answer the following questions.

- 1- Identify user classes of the described system and mark them as primary / secondary [2]
- 2- Write one persona of any primary class you identified [5]
- 3- Write a user scenario with respect to automated attendance and leave management system [5]

Q6. The figure given below is used to apply for leaves at a Pakistani university. The leave type can be medical, casual, or annual. "Date From" and "Date To" shall be entered (typed) with keyboard in the format "dd-mm-yyyy". Remarks is an optional field, and one can write anything like the reason why you want to go on leave. In the "Upload application" one may upload supporting documents if there are any, otherwise it can be left as it is. [5]

Suppose you have to apply for a casual leave from 01-07-2024 to 02-07-2024, and you do not have anything to write in Remarks, or upload in "Upload Application" field, i.e. these fields take default values. Assume your hand is initially on the keyboard, and you move between fields using your mouse. Use KLM to calculate the time to perform this task. Show intermediate working, and follow the format discussed in class.

LEAVE APPLICATION FORM		Op	Sec
Leave Type	Select Leave Type	K	0.28
Date From	Please choose a date	P	1.1